Practice Assignment 1

Scatter Plots, Correlation, and Linear Regression

Discussion on Wednesday, May 6th at 4 PM

This practice assignment will use the **Country_Information.csv** file on GitHub. In this data set, some variables were collected from 41 countries around the world. The variables in the data set are

1. Life expectancy (LifeExpectancy),
2. Average number of years of schooling (YearsSchool),
3. Gross domestic product per capita (GDP) in U.S. dollars,
4. The gender inequality index (GII), which rates the amount of gender inequality in the given country (bigger values mean more inequality),
5. An index score rating the quality of education in the given country (EduIndex), and
6. Public health expenditures, which measures how much the given country spends on public health programs (PubHealthExp).

1) Read the dataset into **R**. Also, if you have not already, install the **math3150package** library from my GitHub (instructions for this are in the notes) and load that package. This will also load the tidyverse.

2) Create a correlation matrix with all of the variables except country and include the correlation matrix here. A screenshot or copy and paste is fine. Feel free to round the correlations to 3 decimal places. You can do this with the **cor()** function.
   a. Which two variables have the strongest linear relationship? Explain briefly how you know.

   b. Which two variables have the weakest linear relationship? Explain briefly how you know.

   c. Suppose you want to predict life expectancy. Which other variable would be most useful in predicting performance? Note that you cannot use life expectancy to predict itself.

   d. Which variable would be least useful in predicting life expectancy?

3) Fit a linear regression model using GII as the explanatory variable and life expectancy as the response variable.
   a. Using ggplot, make a scatterplot of life expectancy (on the y-axis) against GII (on the x-axis). In addition, include the line of best fit by adding this line of code to your ggplot:
   geom_smooth(method = "lm", formula = "y ~ x", se = F)
   Include the scatterplot here.

   b. Using the scatterplot, describe the relationship between the two variables. You may want to use the correlation between them that you found in Question 1 when describing the strength.

   c. Write down (or type) the regression equation **in the context of the problem**. Keep at least 2 decimal places for the slope.

   d. Use the **predict()** function to find the life expectancy for a country that has a GII of 0.5.

e. Use the **predict()** function to find the life expectancy for a country that has a GII of 3.1. Explain why we cannot be confident in that prediction.

f. Interpret the slope of this regression model in the context of the problem.

g. Interpret the intercept of this regression model in the context of the problem.

h. According to the output of the model (when you use the **summary()** function), can we be confident that the true slope is not equal to zero? Explain.

i. Use the **predict()** function to obtain a 95% prediction interval for the life expectancy for any country that has a GII of 0.5. Report the prediction interval and interpret the interval.

4) Now fit a multiple linear regression model using life expectancy as the response variable and GDP, GII, education index, and public health expenditures as explanatory variables.

   a. Write down the multiple regression equation in the context of the problem. You can round all slopes to two decimal places except the one for GDP. Keep 7 decimal places for that one.

   b. Use the regression equation (or **predict()** function) to predict the life expectancy of a country that a GDP of 20,000 per capita, a GII of 0.4, an education index of 0.3, and a public health expenditure measure of 1.8.

   c. Report and **interpret** the coefficient of determination for this model.

   d. Interpret the slope of the education index variable in the context of the problem.

   e. Interpret the intercept of this regression model in the context of the problem.

   f. Using a significance level of 0.05, which (if any) variables in this model are not considered significant? Briefly explain how you know.

   g. Use the **residual_plots()** function in **math3150package** with **resid_type = "raw"** to create a residual and QQ plot for this multiple regression model. Comment on what these plots suggest about the regression assumptions here.

   h. Use the **influence_plots()** function in **math3150package** to create influence plots. Looking at Cook's distance, DfFits, and the DfBetas plots, which points seem to be more influential? What countries are those points?